General Mathematics — Exercise Sheet 6

Standard limits and intermediate values

Variant: Statements

Exercise 1 (A power with alternating small increments). For integers $n \geq 2$, let

$$a_n = \left(1 + \frac{(-1)^n}{n}\right)^n.$$

Find the limits of the even-index and odd-index subsequences. Decide whether $(a_n)_{n\geq 2}$ converges. Use the logarithmic standard limit from Lecture 6 and the subsequence property from Lecture 4.

Exercise 2 (Cancellation beyond an equivalent). Without derivatives or series, evaluate

$$\lim_{x\to 0} \frac{\sqrt{1+x} - 1 - x/2}{x^2}, \qquad x > -1,\ x \neq 0.$$

Explain why replacing $\sqrt{1+x} - 1$ by its equivalent $x/2$ does not justify an answer. Use rationalisation and Lecture 5's limit laws.

Exercise 3 (The same base with different exponents). For nonzero real x, define

$$A(x) = (1+x^2)^{1/x}, \qquad B(x) = (1+x^2)^{1/x^3}.$$

Find the limit of A as $x \to 0$ and the two one-sided limits of B. Decide whether B has a two-sided limit. Use logarithms and explain why all real powers are defined on the whole deleted domain.

Exercise 4 (A root without a polynomial formula). Prove that $2^x + x = 4$ has exactly one real solution. Locate it first in $(1, 2)$ and then in an interval of length $1/2$. Give a midpoint approximation with an error bound from that interval. Use continuity and the order of the school exponential, together with Lecture 6's intermediate value theorem and bisection bound.

Exercise 5 (A product of standard equivalents). Evaluate, as $x \to 0$,

$$F(x) = \frac{[(1+x)^{2/3} - 1]\tan x}{e^{x^2} - 1}.$$

Use the positive-base real power. Give a deleted neighbourhood where the expression is defined, then justify product and quotient replacement by equivalents using Lecture 6.